DeepSeek

Model Config

[huggingface DeepSeek config]

Memory Analysis

Static Memory

matrix	shape	per layer parameter	num layers	Total parameter	Total Bytes(GB)
embedding	[129280, 7168]	926679040	1	926679040	0.86
mla_q_a_proj	[1536, 7168]	11010048	61	671612928	0.63
mla_q_nope	[128*128, 1536]	25165824	61	1535115264	1.43
mla_q_rope	[128*64, 1536]	12582912	61	767557632	0.71

matrix	shape	per layer parameter	num layers	Total parameter	Total Bytes(GB)
mla_k_nope	[512, 7168]	3670016	61	223870976	0.21
mla_k_rope	[64, 7168]	458752	61	27983872	0.03
q_absorb	[128*128, 512]	8388608	61	511705088	0.48
uv_absorb	[128*128, 512]	8388608	61	511705088	0.48
o_proj	[7168, 128*128]	117440512	61	7163871232	6.67
mlp	[7168, 18432]	132120576	3 * 3	1189085184	1.11
moe router	[7168, 2048]	14680064	3 * 256 * 58	653908770816	609
moe shared expert	[7168, 2048]	14680064	3 * 58	851443712	0.79
output linear	[7168, 129280]	926679040	1	926679040	0.86

attention static memory + mlp static memory = 11.75GB

per expert memory = 7168 * 2048 * 3 * 61 = 2.5GB

EP	expert per die	MoE static memory
16	17	42.5GB
32	9	22.5GB
64	5	12.5GB
128	3	7.5GB

Dynamic Memory (only MoE and kv_cache memory is considered.)

moe dynamic memory = ffn_bs * 7168 * 61

kv_cache memory = 2* attn_bs * kv_len *(512+64) * 61 [attn_bs, kv_len, 1, (512+64)]

TODO:

- [ ] attn dynamic memory MLP dynamic memory

A3Pod [64GB per die]

```
attention static memory + mlp static memory + per expert memory * (1 + 256 / EP) + kv_cache memory + moe dynamic memory < 64 * 0.9
```

DeepEP EP16

attn_bs * kv_len < 51187

kv_len=2k, 4k, 8k, 16k, 128k, memory bound

DeepEP EP32

attn_bs * kv_len < 356783.

kv_len=8k, 16k, 128k, memory bound

TPOT Analysis

latency constraint:

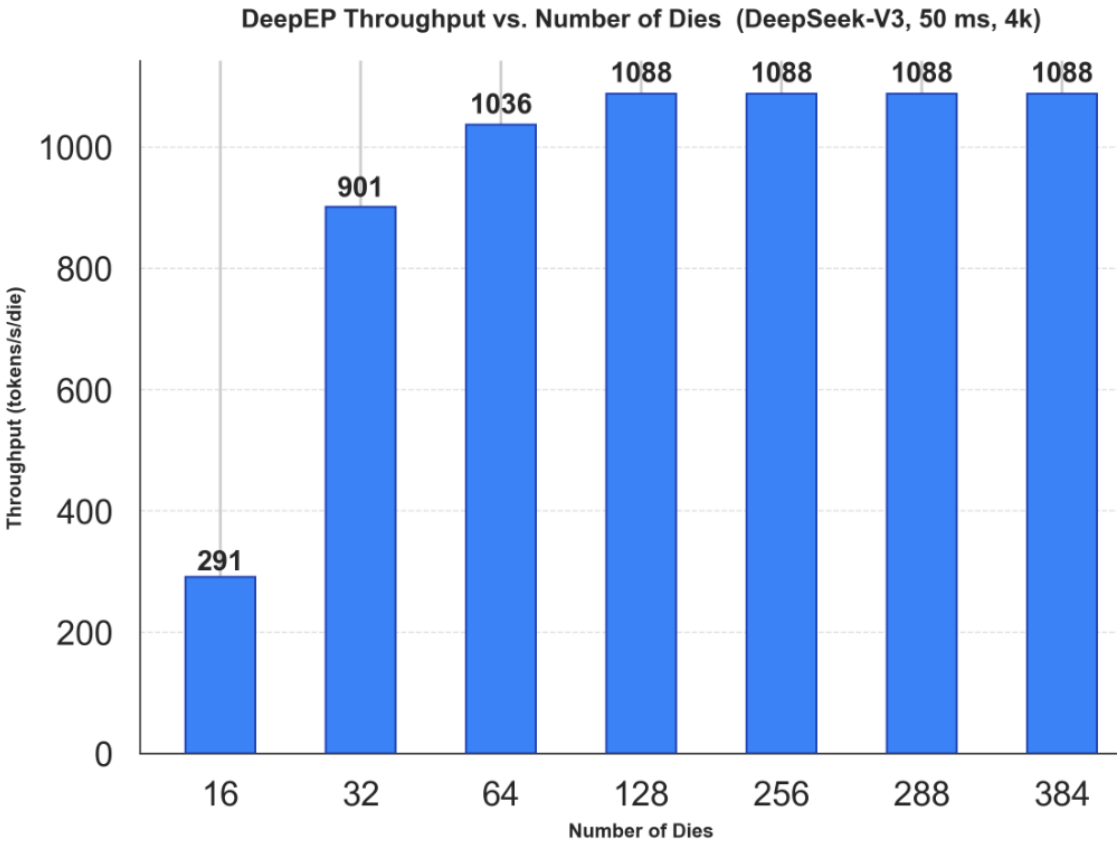
```
attn_time * micro batch num < TPOT / num layers
moe_time * micro batch num < TPOT / num layers
attn_time + moe_time + commu_time < TPOT / num layers
```

TPOT = 20ms

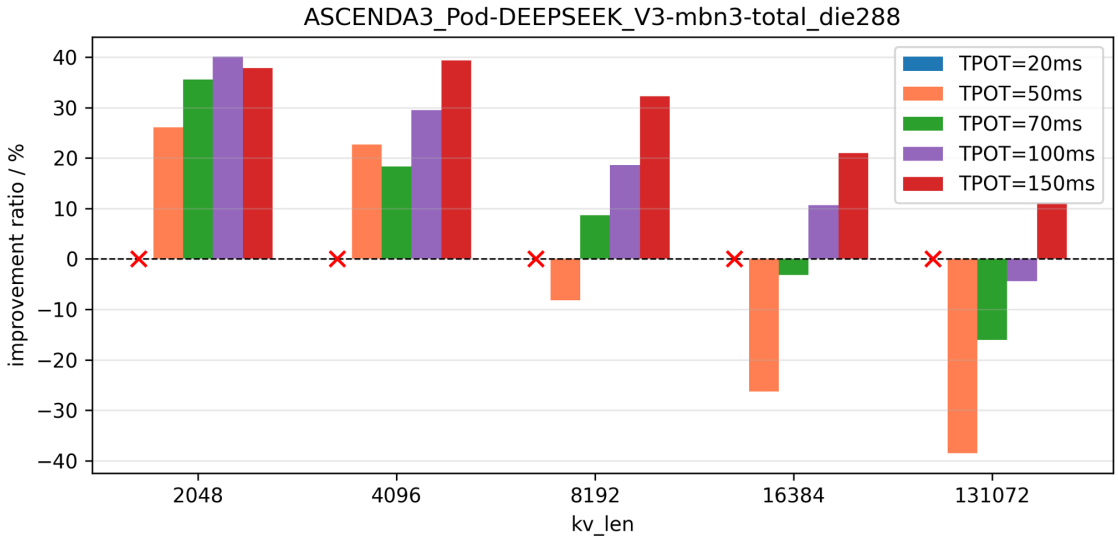
当micro batch num = 3时, attn_time < 20ms * 1.7 / 61 / 3 = 186us, 小于attention访存时间

AFD improvement Analysis

- DeepEP 128die后，吞吐不再增加

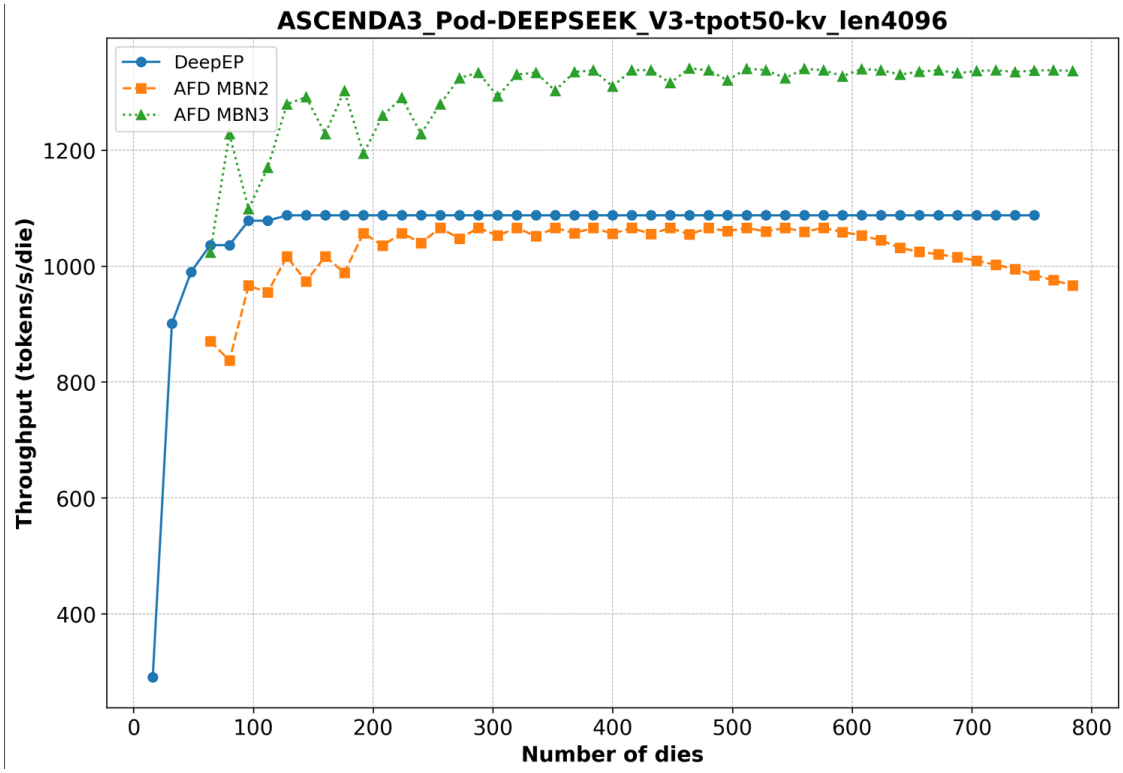


- kv_len增大，收益降低；tpot增大，收益降低

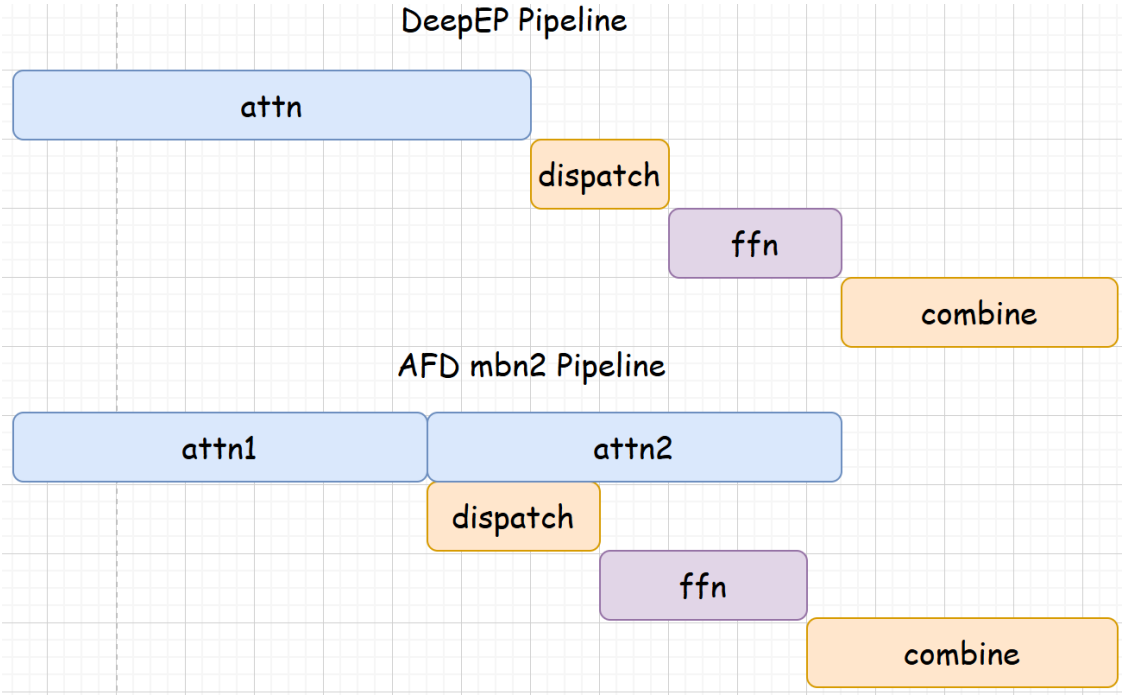


- kv_len增大，attn时间增大，A:F增大，通信开销大。tpot150ms，kv_len128k，A:F可以达到6:1
- kv_len=8k, 16k, 128k，AFD attn batch size低于20，算力利用率极低
- tpot增大，batch size增大，算力利用率高。tpot20ms, 50ms时，DeepEP batch size已经为40左右，算力利用率较高，tpot增大，DeepEP算力利用率增长小于AFD算力利用率增长

- kv_len=4k、tpot=50ms时，mbn2性能劣化



序列长度4k、50ms时延约束时，DeepEP的attn时间与通信+ffn时间相近。



Qwen3-235B-A22B

Model Config

[huggingface Qwen3-235B-A22B config]

Memory Analysis

Static Memory

matrix	shape	per layer parameter	num layers	Total parameter	Total Bytes(GB)
embedding	[151936, 4096]	622329856	1	622329856	0.58
q_proj	[4096, 64*128]	33554432	94	3154116608	2.94
k_proj	[4096, 4*128]	2097152	94	197132288	0.18
v_proj	[4096, 4*128]	2097152	94	197132288	0.18
o_proj	[64*128, 4096]	33554432	94	3154116608	2.94
moe router	[4096, 1536]	6291456	94 * 128 * 3	227096395776	211.5
moe shared expert	[4096, 1536]	6291456	94 * 2 * 3	3548381184	3.3

matrix	shape	per layer parameter	num layers	Total parameter	Total Bytes(GB)
output linear	[151936, 4096]	622329856	1	622329856	0.58

attention static memory = 6.24GB per expert memory = 4096 * 1536 * 3 * 94 = 1.65GB

EP	expert per die	MoE static memory
16	18	29.7GB
32	10	16.5GB
64	6	9.9GB
128	4	6.6GB

Dynamic Memory (only MoE and kv_cache memory is considered.)

moe dynamic memory = ffn_bs * 4096 * 94

kv_cache memory = 2* attn_bs * kv_len *4 * 128 * 94 [attn_bs, kv_len, 4, 128]

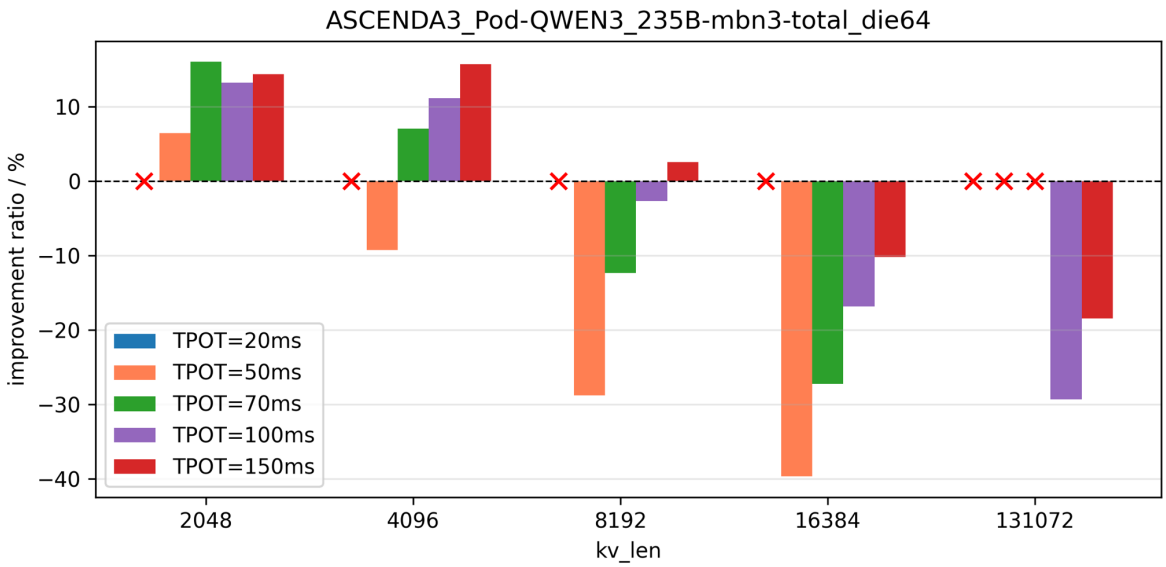
TODO: attn dynamic memory MLP dynamic memory

- A3Pod [64GB per die]

attention static memory + mlp static memory + per expert memory * (1 + 256 / EP) + kv_cache memory + moe dynamic memory < 64 * 0.9

DeepEP EP16, attn_bs * kv_len < 241618. kv_len=8k, 16k, 128k, memory bound

AFD improvement Analysis



与DeepSeek相比

- 层数增多导致bs变小
- FFN访存量降低，A:F较大，通信开销大。DeepSeek A:F为2:1，Qwen235B A:F为4:1
  - 每层路由专家128个，相比于DeepSeekV3减半
  - 每个路由专家[4096, 1536]，相比DeepSeekV3，每个路由专家参数量减少40%
- 模型小，总die数64，也有收益